

Minor in Finance – Semester VI Unit 1 -

Introduction to Fintech





Financial Technology (Fintech)

[\ fə- 'nan(t)-shəl \ tek- 'nä-lə-jē]

New tech that seeks to improve and automate the delivery and use of financial services.

What is Financial Technology – Fintech?



Fintech, the word, is a combination of "financial technology".

Financial technology (Fintech) is used to describe new tech that seeks to improve and automate the delivery and use of financial services.

Key differences between Fintech and Traditional

Banks

	Fintech	Traditional Banking
Accessibility	Fully digital, available 24/7, no physical branches	Limited hours for in-person banking. Some digital offerings.
Services & Offerings	More variety, including HYSA, crypto, and instant transfers.	Comprehensive, including mortgages, loans, checking, and savings.
Costs	Tend to charge lower fees and offer higher interest rates due to lower overhead costs.	Higher fees and lower interest rates, including overdraft fees and maintenance fees.
Customer Support	Primarily chat or email support, often driven by AI. Limited phone support is available; no in-person support is provided.	Offers customer service in person and by phone, often providing a more personalized experience.

Global Fintech Adoption Index

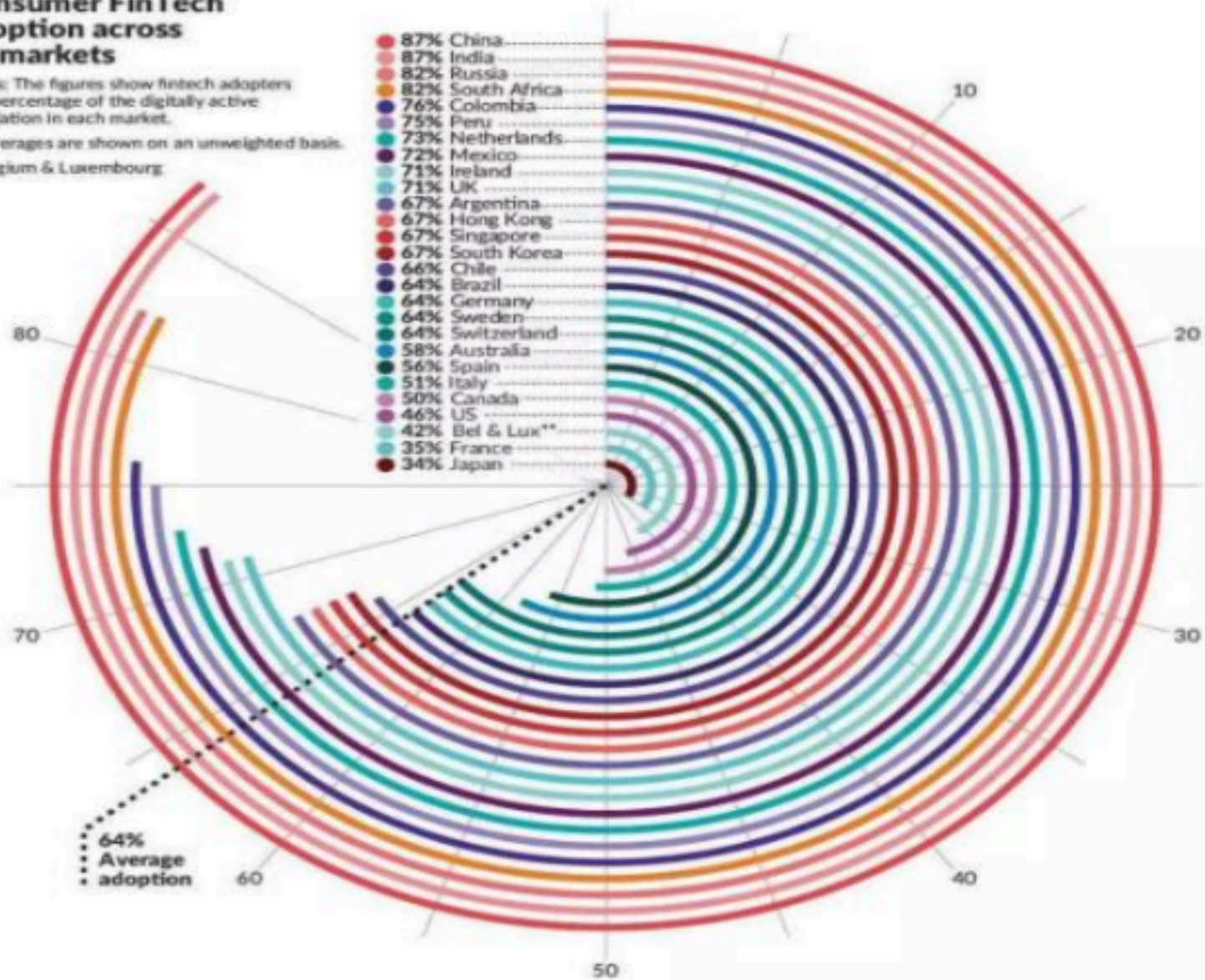
- Global Fintech Adoption Index is used exploring how FinTech has improved and expanded its offerings around the world, spurring change across the financial services industry.
- The Global FinTech Adoption Index has been determined by EY Global Financial Services since 2015.
- Only 27 countries are still involved in determining the index.

Consumer FinTech adoption across 27 markets

Notes: The figures show fintech adopters as a percentage of the digitally active population in each market.

All averages are shown on an unweighted basis.

**Belgium & Luxembourg



What Are Examples of Fintech?

- **Robo-advisors** are apps or online platforms that optimally invest your money automatically, often for little cost, and are accessible to ordinary individuals.
- **Investment apps** like Robinhood make it easy to buy and sell stocks, exchange-traded funds (ETFs), and cryptocurrency from your mobile device, often with little or no commission.
- **Payment apps** like PayPal, Venmo, Block (Square), Zelle, and Cash App make it easy to pay individuals or businesses online and in an instant.
- **Personal finance apps** such as Mint, YNAB, and Quicken Simplifi let you see all of your finances in one place, set budgets, pay bills, and so on.
- **Peer-to-peer (P2P) lending** platforms like Prosper Marketplace, LendingClub, and Upstart allow individuals and small business owners to receive loans from an array of individuals who contribute microloans directly to them.
- **Crypto apps**, including wallets, exchanges, and payment applications, allow you to hold and transact in cryptocurrencies and digital tokens like Bitcoins.
- **Insurtech** is the application of technology specifically to the insurance space. One example would be the use of devices that monitor your driving in order to adjust auto insurance rates.

Fintech Users

Business-to-business (B2B), also called B-to-B, is a form of transaction between businesses. Business-to-business refers to business that is conducted between companies, rather than between a company and individual consumer.

Businesses can easily get loans, financing and other financial services through mobile technology. For example, Credibly, Lendkey, BFS capital.



Fintech Users

B2C - Business to client or B2C

- Cash apps like PayPal, Paytm, Alipay, Apple Pay etc.
- Allow clients or customers to transfer money via the internet or mobile technology.



Fintech Users

Consumer-to-consumer (C2C) fintech services.

This type of service is designed to help consumers interact with each other. One of the best examples of these services is Venmo. Rather than trading cash or checks, consumers can send money directly to each other, instantly and with no charge. Even Google has made inroads in this space by allowing Gmail users to attach cash directly to emails.

Pros and cons of Fintech

Pros of Fintech

- Real-Time Data Collection
- Better Customer Service
- Helps in Decision Making
- Easier Payments
- Effective Interaction

Cons of Fintech

- Lack of Security
- No Uniform Standards
- Complex Networks

Evaluating of Fintech

1866-1967
FinTech **1.0**

1967-2008
FinTech **2.0**

2008-current
FinTech **3.0**

2008-current
FinTech **3.5**

GEOGRAPHY



Global /
Developed



Global /
Developed



Developed



Emerging /
Developing

KEY ELEMENTS



Infrastructure /
computerisation
Developed



Traditional /
Internet



Mobile /
Start-ups /
New entrants



Mobile /
Start-ups /
New entrants

SHIFT ORIGIN



Linkages



Digitalization

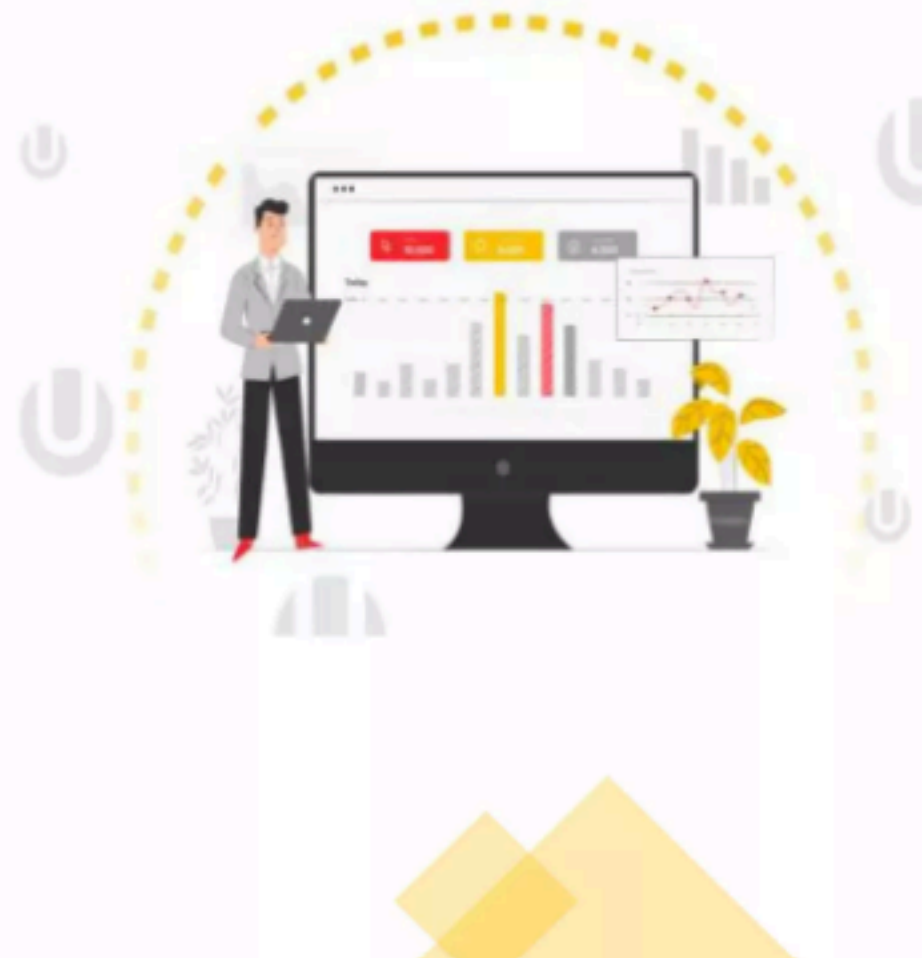


2008 financial crisis /
smartphone



Last mover
advantages

Fintech 1.0 (1886-1967) is about infrastructure. This is an era when we can first start speaking about financial globalization. It started with technologies such as the telegraph as well as railroads and steamships that allowed for the first time rapid transmission of financial information across borders. The key events on this timeline include first transatlantic cable (1866) and Fedwire in the USA (1918), the first electronic fund transfer system, which relied on now-archaic technologies such as the telegraph and Morse code. The 1950s brought us credit cards to ease the burden of carrying cash. First, Diner's Club introduced theirs in 1950, American Express Company followed with their own credit card in 1958.





Fintech 2.0 (1967-2008) is about banks

This period marks the shift from analog to digital and is led by traditional financial institutions. It was the launch of the first handheld calculator and the first ATM installed by Barclays bank that marked the beginning of the modern period of fintech in 1967.

The 1980s saw the rise of bank mainframe computers and the world is introduced to online banking, which flourished in 1990s with the Internet and e-commerce business models. Online banking brought about a major shift in how people perceived money & their relationship with financial institutions. This era ends with the Global Financial Crisis in 2008

Fintech 3.0 (2008-Current) is about start-ups.

Important factor that shaped the face of fintech is the mass-market penetration of smartphones that has enabled internet access for millions of people across the globe. Smartphone has also become the primary means by which people access the internet and use different financial services. 2011 saw the introduction of Google Wallet, followed by Apple pay in 2014.



FinTechs and their evolution in Indian Banking Context

- FinTech's evolution in the Indian banking context has been **transformative**, driving innovation, enhancing customer experience, and reshaping the financial services landscape.
- Over the past decade, India has emerged as one of the **largest and fastest growing FinTech markets** globally, largely fuelled by advancements in technology, increasing smartphone penetration, and supportive government policies.
- Here's an overview of the **key stages and trends in the evolution of FinTech within Indian banking**:

1. Initial Phase (2008-2014):

The Emergence of FinTech in Payments

❖**Digital Payments:** The early years of FinTech in India were largely centred around digital payments. Companies like Paytm, Mobikwik, and Oxigen pioneered mobile wallets and prepaid payment instruments (PPIs), providing an alternative to traditional banking payment methods.

❖**E-Commerce Integration:** The rise of ecommerce platforms such as Flipkart and Snapdeal created a demand for seamless digital payment solutions, which FinTech firms quickly filled.

❖**Payment Gateways:** Players like Razorpay and CCAvenue introduced easy-to-integrate payment gateway solutions, enabling small businesses and startups to accept digital payments.

2. Growth Phase (2015-2017):

Rise of FinTech in Lending and Regulatory Push

❖**Government Initiatives:** The Indian government launched several initiatives such as **Digital India, Jan Dhan Yojana, and Pradhan Mantri Jan Dhan Yojana (PMJDY)** to foster financial inclusion, giving FinTech companies opportunities to provide digital banking services to the

unbanked and underbanked population.

- ❖ **UPI Revolution:** The introduction of **Unified Payments Interface (UPI)** in **2016** by the National Payments Corporation of India (NPCI) revolutionized the Indian payment system. UPI enabled realtime, mobile based interbank transactions, facilitating the growth of apps like Google Pay, PhonePe, and BharatPe.
- ❖ **FinTech Lending:** Platforms like LendingKart, Capital Float, and Rupeek emerged, offering quick, unsecured loans to MSMEs, and filling the credit gap left by traditional banks. Peer-to peer (P2P) lending platforms like Faircent and Lendbox also began providing alternative financing solutions.

3. Maturity Phase (2018-Present):

FinTech Diversification and Deepening Integration with Banks

- ❖ **Neo banking:** FinTechs like NiYo, RazorpayX, and Jupiter launched neo banking platforms, providing a fully digital banking experience without traditional branches. These platforms offer services such as instant account opening, expense tracking, and automated savings, targeting millennials and digital first consumers.
- ❖ **Digital Lending Platforms:** FinTechs in the digital lending space, such as ZestMoney, Kissht, and FlexiLoans, evolved further by using AI and big data to assess creditworthiness through alternative data sources, making lending more accessible to consumers and small businesses.

❖ **Partnerships with Banks:** Banks recognized the potential of FinTech and began partnering with these firms to cocreate solutions. API banking, for instance, became a cornerstone of such collaborations. Banks like HDFC Bank, ICICI Bank, and Axis Bank now work closely with FinTechs to offer enhanced customer experiences through mobile apps and digital platforms.

❖ **WealthTech and InsurTech:** FinTech also expanded into wealth management and insurance with companies like Groww, Zerodha, PolicyBazaar, and ETMoney offering simplified investment and insurance solutions to the masses.

❖ **Buy Now, Pay Later (BNPL):** The BNPL model, popularized by companies like LazyPay, Simpl, and ZestMoney, became a key feature in the Indian FinTech ecosystem, offering short term credit for ecommerce purchases.

4. Key Drivers of Evolution

❖ **Aadhaar and eKYC:** The development of the Aadhaar system and the integration of eKYC (electronic Know Your Customer) processes played a pivotal role in streamlining digital onboarding for customers. This significantly reduced the friction in opening bank accounts, applying for loans, and getting insurance policies.

❖ **Regulatory Sandbox:** The Reserve Bank of India (RBI) has been proactive in creating a regulatory sandbox, allowing FinTechs to test innovative products in a controlled environment. This has fostered innovation while ensuring consumer protection and financial stability.

❖ **Bharat Bill Payment System (BBPS):** The BBPS has streamlined the bill payment ecosystem, allowing FinTech

players to expand into utility payments and recurring payment solutions for a broad base of consumers.

❖**Open Banking and Account Aggregators:** The open banking initiative, supported by RBI's Account Aggregator framework, has enabled consumers to share financial information securely across banks and FinTechs. This has opened up new opportunities for personal finance management and credit assessment.

5. Recent Trends and Future Outlook

❖**Embedded Finance:** Embedded finance is the next frontier, where FinTech services like payments, lending, or insurance are embedded within nonfinancial platforms, such as ecommerce or travel apps, offering seamless financial solutions.

❖**Blockchain and Digital Currency:** With the growing interest in blockchain and the RBI working on a Central Bank Digital Currency (CBDC), the future of FinTech in India may also involve decentralized financial services, cross border payments, and digital currencies.

❖**AI and Data Analytics:** FinTech companies are increasingly leveraging artificial intelligence and big data analytics to enhance fraud detection, automate credit scoring, and personalize financial services.

❖**Financial Inclusion:** The focus on financial inclusion remains a priority, with FinTechs creating innovative products tailored for rural areas, agricultural finance, and MSMEs, helping bridge the gap between formal banking and underserved communities.

How Banks are coping up with this disruption/changes/shifts

Adaptation by Banks in evolving environment

Banks in India are adapting to the rapid changes brought by FinTech innovation in various ways. While FinTech firms have introduced **agility, convenience, and new business models**, banks have responded by embracing digital transformation, collaborating with FinTechs, and developing inhouse innovations to stay competitive. Here's how Indian banks are coping with these shifts:

1. Collaborations and Partnerships with FinTechs

❖**API Banking and Open Banking:** Banks are partnering with FinTech companies to offer Application Programming

Interfaces (APIs), enabling third party developers to build on banking services. For example, ICICI, HDFC, and Yes Bank have embraced open banking models.

- ❖ **Strategic FinTech Partnerships:** Banks form alliances with FinTechs to leverage technological expertise, such as Axis Bank partnering with Freecharge and SBI working with digital lending platforms.

Adaptation by Banks in evolving environment

2. Digital Transformation and Automation

- ❖ **Digital First Banking:** Banks are offering seamless digital services, such as Kotak Mahindra Bank's 811 and ICICI's iMobile, with full functionalities online.
- ❖ **AI and Automation:** Investments in AI-driven chatbots (like SBI's SIA) and RPA are enhancing customer service and operational efficiency.
- ❖ **Paperless Onboarding and e-KYC:** Banks use e-KYC and Aadhaar based digital onboarding to simplify customer acquisition.

3. Development of Neo Banking and Digital Only Platforms

- ❖ **In House Neo Banks:** Banks like ICICI and Axis Bank have launched digital only platforms such as InstaBIZ and Burgundy for tech savvy customers.

- ❖ **Acquisitions and Investments:** Banks are investing in FinTech start ups, such as HDFC Bank investing in payment and lending FinTechs.

Adaptation by Banks in evolving environment

4. Enhancing Digital Payment Ecosystem

- ❖ **UPI Integration:** Banks are heavily involved in the UPI ecosystem, partnering with payment apps like Google Pay and PhonePe.
- ❖ **Contactless Payments:** Banks are promoting NFC based contactless payments, like HDFC Bank's PayZapp and Axis Bank's Scan & Pay.

5. Adopting FinTech Driven Lending Models

- ❖ **Digital Lending and Instant Loans:** Banks adopt FinTech models for quick digital lending to MSMEs, partnering with platforms like Capital Float and Lendingkart.
- ❖ **Buy Now, Pay Later (BNPL):** Banks offer BNPL services, with players like ICICI Bank and HDFC Bank introducing their own versions.

Adaptation by Banks in evolving environment

6. Investing in Cybersecurity and Data Protection

- ❖ **Strengthening Security:** Banks invest in advanced data encryption, biometric authentication, and multifactor authentication (MFA) to secure digital services.
- ❖ **Fraud Detection Using AI:** AI and machine learning are used for real time fraud detection, helping banks identify unusual activities quickly.

7. Building Innovation Labs and Incubators

- ❖ **Innovation Hubs:** Banks like ICICI and Axis Bank have set up innovation labs focused on blockchain, AI, and API banking to cocreate solutions with FinTechs.
- ❖ **Hackathons:** Banks host hackathons to collaborate with FinTech developers, spurring new ideas and rapid technology adoption.

Adaptation by Banks in evolving environment

8. Wealth Tech and Robo-Advisory Services

- ❖ **Digital Wealth Management:** Banks compete with Wealth Tech platforms by offering end-to-end digital trading through apps like ICICI Direct and HDFC Securities.
- ❖ **Robo-Advisors:** AI-based robo advisors help customers make personalized investment decisions.

9. Regulatory Adaptation and Sandbox Participation

- ❖ **Regulatory Sandbox:** Banks participate in the RBI's regulatory sandbox, testing new technologies like blockchain and AI in controlled environments.
- ❖ **Account Aggregator Framework:** Banks adopt the Account Aggregator (AA) system, allowing secure data sharing for better credit assessment.

Adaptation by Banks in evolving environment

10. Customer-Centric Product Development

- ❖ **Personalized Digital Products:** Banks use big data analytics to offer customized banking products and targeted offers based on customer behaviour.
- ❖ **Omnichannel Experience:** Banks integrate physical and digital channels for an omnichannel experience, ensuring seamless transitions between platforms.

Conclusion

Banks in India have effectively responded to FinTech disruption by embracing digital transformation, collaborating with FinTechs, and leveraging cutting-edge technologies.

While FinTechs continue to push the boundaries of innovation, banks are integrating these advancements into their core operations, driving a symbiotic relationship that is reshaping the financial ecosystem.

Traditional banks that **adapt and innovate** will **continue to thrive** in this evolving landscape.

Impact of Fintech on Startups and Emerging Markets

For Startups:

- **Access to Capital:** FinTech has catalyzed alternative funding solutions like peer-to-peer (P2P) lending and crowdfunding, allowing startups to raise capital without relying solely on traditional banking institutions. This is crucial for new ventures that may lack a formal credit history or collateral.

- **Reduced Operational Costs:** FinTech solutions help startups to automate financial operations, manage processes more efficiently, and lower the costs associated with traditional banking services, making them more competitive.
- **Faster, Data-Driven Decisions:** Startups can leverage artificial intelligence (AI) and machine learning (ML) in FinTech applications to analyze vast datasets, assess risk more accurately, and make informed business decisions quickly.
- **Global Connectivity:** FinTech facilitates seamless cross-border payments and commercial transactions, helping startups in emerging markets connect with global suppliers and customers more easily.

For Emerging Markets:

- **Enhanced Financial Inclusion:** FinTech has been a major driver of financial inclusion in regions with underdeveloped traditional banking infrastructure. Mobile money platforms (e.g., M-Pesa) and digital payment systems are bringing basic financial services to previously unbanked or underbanked populations in rural and remote areas.
- **Economic Growth:** The expansion of financial access contributes to broader economic development by enabling individuals to save, invest, and manage financial risks more effectively. This, in turn, boosts entrepreneurial activity and improves household welfare.

- **Formalization of the Economy:** Accepting digital payments helps integrate small businesses and informal traders into the formal economy, which can lead to better transaction recording and access to formal credit in the future.
- **Infrastructure Leapfrogging:** Emerging markets can "leapfrog" traditional banking infrastructure by going straight to mobile-based solutions, which require less physical investment.

Fintech Typology

- The typology of Financial Technology (**FinTech**) is broad, encompassing various sectors, technologies, and business models that use software and modern technology to provide or enhance financial services.
- It is generally categorized based on the services provided or the underlying technology used.

Categories by Service/Business Model

Most commonly, FinTech is classified by the specific financial services it delivers to consumers and businesses:

- **Digital Payments and Money Transfers:** This category includes apps and services that facilitate online, mobile, and peer-to-peer (P2P) transactions, enabling fast and often cheaper money transfers than traditional banking methods. Examples include mobile wallets and point-of-sale systems.
- **Lending and Alternative Financing:** These platforms offer alternative credit options such as P2P lending, crowdfunding, microloans, and "Buy Now, Pay Later" (BNPL) services. They often use AI for faster, more efficient credit scoring and loan approvals.
- **Digital Banking (Neobanks):** These are online-only financial institutions that operate without physical branches. They provide traditional banking services like checking and savings accounts, often with lower fees and more user-friendly interfaces.
- **WealthTech/Investment and Wealth Management:** These services use technology to automate and democratize investing and portfolio management. This includes robo-advisors that offer algorithmic investment advice and commission-free trading platforms.
- **Insurtech (Insurance Technology):** This segment uses data analytics, AI, and the Internet of Things (IoT) to streamline the insurance industry, from policy underwriting and risk assessment to claims processing.
- **Personal Finance Management (PFM):** PFM tools include budgeting apps, expense trackers, and credit score monitoring services designed to help individuals manage their

personal finances more effectively.

- **RegTech (Regulatory Technology):** These companies provide automated solutions to help financial institutions comply with regulations efficiently. Services include identity verification (KYC), transaction monitoring, and automated reporting.

Categories by Core Technology

FinTech can also be viewed through the lens of the core technologies that enable these services:

- **Blockchain and Cryptocurrency:** Utilizing decentralized, distributed ledger technology for secure and transparent transactions, including the use of Bitcoin, Ethereum, and Decentralized Finance (DeFi) applications.
- **Artificial Intelligence (AI) and Machine Learning (ML):** Used for data analysis, fraud detection, personalized financial advice, and enhancing decision-making processes like loan underwriting.
- **Cloud Computing:** Provides the scalable and agile infrastructure necessary for FinTech companies to host applications and store data securely and cost-effectively.
- **Big Data and Data Analytics:** The use of large datasets to understand customer behavior, assess risk more accurately, and offer personalized products and services.

- **Application Programming Interfaces (APIs):** The fundamental "glue" that allows different financial systems and third-party applications to connect and share data securely, a key component of the open banking movement.

OVERVIEW OF INDIA'S APPROACH TO REGULATING FINANCIAL SERVICES

- India's regulatory framework for financial services reflects a distinctive integration of technology, targeted policy actions, and market-driven opportunities that align with the country's specific needs. The Indian Financial Technology ("**FinTech**") sector has emerged as a global leader, ranking third among the world's fastest-growing FinTech markets based on investment flow, deal activity, and overall market robustness.
- As of 2024, India's FinTech market is valued at approximately USD 110 billion and is projected to grow to around USD 420 billion by 2029, reflecting a robust compound annual growth rate ("**CAGR**") of 31%. With over 9,000 FinTech companies, India holds the third-largest ecosystem globally in terms of the number of FinTech entities and accounts for 14% of the nation's total startup funding.
- The country also leads in consumer adoption, with a FinTech adoption rate of 87% – significantly surpassing the global average of 67%.

- To keep pace with the rapidly evolving innovations introduced by FinTechs, financial regulators in India are closely tracking developments across the sector. They are embracing a more agile, tech-oriented mindset and moving beyond traditional institution-focused oversight to adopt a broader, activity-based regulatory approach.

KEY DEVELOPMENTS IN THE INDIAN FINTECH SECTOR

- **RBI FinTech department:** In January 2022, the RBI established a FinTech department to identify innovation opportunities associated with the FinTech sector. During the financial year 2023–24, the department:
 - a. conducted multiple pilot programs involving various use cases of central bank digital currencies, India's own digital currency;
 - b. organized several events under India's G20 Initiative; and c. set up more than 75 digital banking units in over 75 Indian districts.

Additionally, the department addresses emerging challenges such as cybersecurity threats and data privacy concerns, ensuring that technological progress aligns with robust regulatory standards.

KEY DEVELOPMENTS IN THE INDIAN FINTECH SECTOR

- **Framework for Self-Regulatory Organization in the FinTech sector (“Self-Reg Framework”):**

- The RBI has in May 2024 issued a Self-Reg Framework, to establish a structured and collaborative regulatory approach to manage the rapidly evolving FinTech landscape.
- This framework outlines the process of recognition of self-regulatory organizations as well as their broad functions and governance standards.

- By empowering the self-regulatory organizations with the responsibility to oversee their members, the RBI seeks to enhance transparency, accountability, integrity, fairness, consumer protection, and promote fair competition.

KEY DEVELOPMENTS IN THE INDIAN FINTECH SECTOR

- **Digital payment systems for persons with disabilities:**

- On October 11, 2024, the RBI issued guidelines advising payment system participants (“**System Participants**”) to review payment systems and devices from the perspective of accessibility by persons with disabilities and carry out necessary modifications that may be required.
- The guidelines have clarified that System Participants must ensure that the modifications do not compromise security aspects of their systems.

KEY DEVELOPMENTS IN THE INDIAN FINTECH SECTOR

- **Framework for Responsible and Ethical Enablement of Artificial Intelligence (“FREE-AI”) in the financial sector:**

- As part of the Statement on Developmental and Regulatory Policies released alongside the Monetary Policy Statement on December 6, 2024, the RBI announced the formation of a dedicated committee to develop a FREE-AI in the financial sector.
- The initiative reflects the RBI’s proactive approach toward ensuring that the integration of Artificial Intelligence (“AI”) into financial services is both ethical and responsible.
- The FinTech Department at the RBI’s Central Office has been designated to provide secretarial support to the committee.

KEY DEVELOPMENTS IN THE INDIAN FINTECH SECTOR

- Earlier this year, the RBI released a ‘*Statement on Developmental and Regulatory Policies*’ outlining key policy measures across areas such as payment systems and FinTech.

These include:

- Proposed guidelines to ensure responsible lending practices, particularly concerning the imposition of foreclosure charges or pre-payment penalties on loans;
- The release of a discussion paper focused on exploring new capital-raising avenues for primary (urban) co-operative banks; and the creation of the Reserve Bank Climate Risk Information System (“**RB-CRIS**”), which is intended to enhance the assessment and management of climate-related financial risks, etc.
- These measures reflect the RBI’s continued focus on promoting transparency, financial stability, and sustainable practices within the Indian financial sector.

REGULATORY BODIES

- **The RBI:** The RBI is India's central bank and is responsible not only for securing monetary stability, but also to operate the currency and credit system in India and a modern monetary policy. The RBI is the regulatory body for the banking and credit system, payment systems, as well as foreign exchange. Essentially, it is responsible for regulating money market and foreign exchange market in India. The RBI regulates banks, NBFCs, payment system providers, and credit bureaus. For the purposes of governance, the RBI issues regulations, directions, circulars and guidelines for the orderly regulation of the offerings based on banking, credit and payment system products. As the regulator, the RBI is also responsible for granting of licenses and authorizations to the industry players.
- **The Securities and Exchange Board of India (“SEBI”):** The SEBI regulates capital markets, both primary and secondary, in India. It exercises supervisory powers over stockbrokers and investment advisors for their broking and advisory services.
- **The IRDAI:** The IRDAI is the regulatory authority for the insurance sector and insurance products, and regulates insurers, corporate agents, web aggregators and third-party insurance agents, including corporate agents, etc.

REGULATORY BODIES

- **FIU-IND:** The FIU-IND is the national agency responsible for processing and disseminating information relating to suspect financial transactions and coordinating national and international agencies in pursuing the global efforts against money laundering and financing of terrorism. While the FIU-IND is not a sectoral regulator *per se*, over the past year, it has emerged as a key authority in monitoring and regulating virtual digital asset businesses in India in its effort to ensure that these businesses are at all times in compliance with anti-money laundering laws in India.
- Financial Intelligence Unit-India (FIU-IND)
- <https://fiuindia.gov.in>
- Given that Indian regulators are now adopting an activity-based approach to regulation, it is not uncommon for a FinTech to be regulated by different financial services regulators at the same time, depending upon the varied range of products and services offered.

REGULATORY AND INSURANCE TECHNOLOGY

- **The ‘PRAVAAH’ portal:** The e-PRAVAAH portal is likely to serve as a significant regulatory technology introduced by the RBI that will function as a comprehensive digital platform designed to streamline the processes for obtaining regulatory authorizations, licenses and/or approvals.
- This ‘e-PRAVAAH’ portal will streamline submission of numerous application forms covering different regulatory and supervisory departments of RBI, resulting in reduction of the administrative burden on the RBI and other financial institutions.
- This portal embodies the RBI’s commitment to leveraging technology for enhancing transparency, efficiency, and ease of compliance in the financial sector. It is expected to minimize redundancies and errors, ensuring timely and accurate data submission.
- The RBI has advised all applicants, including regulated entities, to use the PRAVAAH Portal for submitting applications for regulatory authorizations, licenses, and approvals to the RBI with effect from May 1, 2025.

REGULATORY AND INSURANCE TECHNOLOGY

- **RBI Retail Direct portal:** This portal has been developed by the RBI to enable retail investors to transact in government securities in a seamless and convenient manner. The portal has recorded 2,06,732 registrations as of July 1, 2024.
- **FinTech Repository:** The ‘FinTech Repository’ is yet another significant initiative of the RBI. This serves as a comprehensive database of FinTech developments and innovations in the country. The repository has been designed with the objective of fostering collaboration and knowledge sharing among various stakeholders resulting in a better understanding of the sector from a regulatory perspective. This could facilitate designing of appropriate policy approaches, especially for adoption of emerging technologies (like AI, machine learning, big data, cloud computing, etc.).

REGULATORY AND INSURANCE TECHNOLOGY

- **The ‘Finquery’ Initiative:** The RBI has recently announced ‘Finquery’, a one of its kind initiative to facilitate direct engagement between FinTech companies and the RBI itself. The ‘Finquery’ portal will be operational for certain hours in its endeavor to seek streamlining of communication and

addressing regulatory concerns of the FinTech sector. This is an important step towards the RBI's dedication to fostering growth and development of the FinTech sector by promoting transparency and providing an opportunity of direct guidance by the regulator.

- **InsurTech:** In India, there have also been certain measures taken by the insurance regulator, i.e., the Insurance Regulatory and Development Authority of India (“**IRDAI**”), to promote and propel technology in the insurance space. In the Insurtech space, three technology driven products have been envisioned under the collective branding ‘*Bima Trinity*’, where ‘*Bima*’ in the Hindi language means ‘insurance’. These include: ‘*Bima Sugam*’, a digital platform proposed to function as the electronic marketplace for insurance; ‘*Bima Vistaar*’, an insurance offering/product proposed for providing basic social safety net cover with combined features of life, health, personal accident and property insurances; and ‘*Bima Vahak*’, a women-centric distribution channel for insurance products. The IRDAI has even issued regulations for ‘*Bima Sugam*’, which is established as an insurance e-marketplace for buying, selling and servicing insurance policies and settlement of insurance claims

REGULATORY AND INSURANCE TECHNOLOGY

- **MuleHunter.AI:** The Reserve Bank Innovation Hub TM (“**RBIH**”) is piloting an AI/Machine Learning (“**ML**”)-powered solution known as MuleHunter.AI , aimed at the efficient identification of mule bank accounts.

- This model has shown promising results in preliminary trials conducted with two major public sector banks.
- To enhance its effectiveness and scale its impact, banks are being encouraged to partner with RBIH in advancing the MuleHunter.AI initiative as a proactive measure to combat financial fraud involving mule accounts

MAJOR OPPORTUNITIES AND CHALLENGES FOR FINTECH IN INDIA

- In India, technology has played a pivotal role in fostering growth of the digital economy. Over time, Indian banks and financial services providers have progressively embraced technology to enhance their outreach, customer service, and operational efficiency, adapting to evolving market conditions and technological advancements.
- Traditionally, banks have been the primary providers of payment services in India. However, rapid technological advancements have disrupted this monopoly and now non-bank entities are seen competing with banks, serving

either as technology service providers to banks or directly offering innovative digital payment solutions.

- The regulatory framework in India has also supported and encouraged greater participation from non-bank entities in the payments space.

MAJOR OPPORTUNITIES AND CHALLENGES FOR FINTECH IN INDIA

- Over the years, the RBI has encouraged the use of electronic payments to meet its objective of a ‘cash-less’ society. This gained enormous momentum during the COVID-19 pandemic, where a massive shift in consumer behavior in the realm of payments was also seen, resulting in immense popularity of contactless payment methods.
- FinTech companies promptly introduced solutions like mobile payment applications, digital wallets, and quick response (“QR”) code payments using Unified Payments Interface (“UPI”) to facilitate contactless payments transactions.
- As a result, India witnessed a tremendous growth in the digital payment volume, from 8.6

billion in the month of May 2022 to 16.9 billion transactions in the month of May 2024.

MAJOR OPPORTUNITIES AND CHALLENGES FOR FINTECH IN INDIA

- Resultantly, digital financial services are now being adopted more widely than ever before. This trend is likely to continue. Given the immediate availability and convenience offered to consumers, digital financial services are likely to continue driving the Indian FinTech sector's growth.
- Further, there has been increased adoption of the 'Account Aggregator' mechanism introduced for secure and seamless sharing of financial information held by an entity regulated by a financial services regulator (i.e., RBI, securities regulator or insurance regulator), of an individual.
- In one of the most basic use cases of this, India has witnessed various innovative FinTech offerings being made available by financial information users that consolidate individuals' financial information and transactions across various kinds of financial services, including bank account, securities, mutual funds and insurance policies, at a single digital platform.

Challenges Facing the Indian Fintech Sector

Despite rapid growth, the fintech industry in India must navigate several significant challenges:

- **Cybersecurity & Data Privacy:** Digitization makes systems vulnerable to cyberattacks and data breaches. Ensuring robust security and clear data protection laws remains a critical concern for both companies and consumers.

- **Regulatory Uncertainty:** The rapidly evolving nature of fintech makes it difficult to formulate a single, comprehensive regulatory approach. The different regulatory bodies (RBI, SEBI, IRDAI) must constantly adapt their frameworks, which can create compliance difficulties for startups.

- **Digital Literacy & Financial Inclusion Gaps:** While urban adoption is high, a significant portion of the rural population lacks digital literacy and trust in online financial services, preferring traditional cash transactions.

- **Infrastructure Deficiencies:** Poor internet connectivity and an inconsistent power supply in remote areas hinder the expansion of digital financial services to the "last mile".

Challenges Facing the Indian Fintech Sector

- **Talent Acquisition & Retention:** The high demand for skilled technology professionals creates a competitive market, making it challenging for smaller startups to attract and retain top talent.
- **Building Sustainable Business Models:** The crowded market, especially in the payments sector, often leads to low margins and intense competition, challenging firms to develop viable, long-term business models that do not heavily burden consumers with extra fees.
- **Intense Competition & Market Saturation:** The market is crowded with numerous startups and large established tech companies, leading to intense competition, price wars, and reduced margins in key segments like digital payments. This saturation makes it challenging for smaller players to differentiate their offerings and achieve sustainable profitability.
- **Access to Capital & Funding:** Early-stage fintech startups often struggle to secure adequate funding, particularly given the increased caution among investors (a "funding winter") that began in 2022. Investors now demand detailed profitability roadmaps, making it difficult for new ventures to raise necessary capital for scaling operations and R&D.

Group Discussion – Business Plan for

Developing a Fintech App